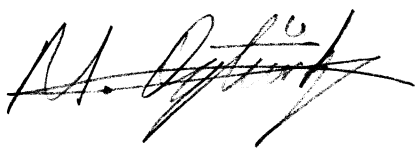
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"On my honor, I have neither given nor received any unauthorised and/or inappropiate asistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."



1. Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}.$$

a) Find the unknown entries α, β, γ, k and $m = ?$

b) Find the eliminated upper triangle matrix $U = ?$

c) Determine the remaining elimination matrice(s)?

2. A geometry in "k" dimensional space is given by $x = 5y$ expression, where 'x' and 'y' represent the axis. The intersection with another geometry gives a volume.

a)

b)

c)

3. Suppose u is a unit vector $\mathbb{R}n$. The problem is about the n by n symetric matrix $H = I - 2u(uT)$

a)

b)

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c)

d)

4. Two tanks are connected, first tank has $v(0) = 10$ liters liquid and the second tank has $w(0) = 40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v - w$.

a) $v(t) = 10 - t + t^2$ $\quad w(t) = 40 - t^2 + t$ $\quad v'(t) = w'(t)$

$v'(t) = 10 - 1 + 2t$ $\quad w'(t) = 40 - 2t + 1$ $\quad 9 + 2t = 41 - 2t$

$= 9 + 2t$ $\quad = 41 - 2t$ $\quad 4t = 32$

$t = 8 \rightarrow$ equal

so our equation set $= \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$

b)

$v'(t) = 9 + 2t$ $\quad v'(1) = 9 + 2 = 11 \rightarrow$ total mixture per.

liquid ↙ ↘ water

$\frac{2}{11} = \%18{,}18$ water

$\frac{9}{11} = \%81{,}82$ liquid

c) $v'(8) = 9 + 2 \cdot 8 = \underline{25}$

$w'(8) = 41 - 2 \cdot 8 = \underline{25}$

} Tank 1 and tank 2 have 25 liters liquid when t is equal to 8.